



Project Overview Node

MTAU Bank is a premium, dark-themed, glassmorphic simulation banking portal designed for final year project demonstrations. It encompasses fiat wire routing, EMV hardware card simulations, fractional asset investments, a dynamic live trading terminal with WebSockets, automated loyalty point distributions, an AI assistant chatbot, and a central administration dashboard.

1. SYSTEM ARCHITECTURE & TECH STACK

The system is built on a modular, lightweight structure designed for low server latency and responsive client interactions:

Ecosystem Layer	Technology Component	Functional Purpose
Frontend UI/UX	HTML5, CSS3 Variables, ES6 JavaScript, Google Font Outfit, Lucide Vector Icons	Implements a premium purple glassmorphic layout, cohesive variables, input states, and responsive navigation grids.
Dynamic Interactive Background	HTML5 Canvas API, physics particle node mapping (<code>cooleffects-lite.js</code>)	Renders connected mauve particles floating upward that gravitate toward the mouse cursor on hover. Disabled on screens < 768px.
Client-side Charting	TradingView Advanced HTML5 Widget API, Chart.js	Streams real-time candlestick charts on the trading desk and plots client account deposit/withdrawal ratios.
Backend Controller	PHP 8.x, Session State management	Handles CSRF token checks, POST transaction logic, administrative overrides, and routes.
Database Layer	MySQL Relational Registry	Manages tables for users, transactions, loans, utility bill records, mutual funds, insurance policies, support desk tickets, saved beneficiaries, and trading logs.
Live Market Feeds	Binance WebSocket API (Crypto) & Open ExchangeRate API (Forex)	Streams live asset rates directly inside the client browser. Utilizes a 15-second caching strategy.

2. RELATIONAL DATABASE SCHEMA (DDL)

The project uses a relational schema. Below is the structure from `mtau_bank.sql.txt`, including columns, keys, and relational triggers:

2.1 Users Registry Table

```
CREATE TABLE users (  
    id INT AUTO_INCREMENT PRIMARY KEY,  
    roll_no VARCHAR(20) UNIQUE NOT NULL,  
    account_class ENUM('Student', 'Business', 'Standard') NOT NULL,  
    account_type ENUM('Current', 'Saving') NOT NULL,  
    full_name VARCHAR(100) NOT NULL,  
    age INT NOT NULL,  
    email VARCHAR(100) UNIQUE NOT NULL,  
    phone VARCHAR(20) NOT NULL,  
    cnic VARCHAR(15) NOT NULL,  
    password VARCHAR(255) NOT NULL,  
    account_no VARCHAR(20) UNIQUE NOT NULL,  
    balance DECIMAL(15, 2) DEFAULT 25000.00,  
    loyalty_points INT DEFAULT 100,  
    card_status ENUM('not_applied', 'issued') DEFAULT 'not_applied',  
    card_tier VARCHAR(30) DEFAULT 'none',  
    card_pin VARCHAR(4) DEFAULT NULL,  
    card_blocked TINYINT(1) DEFAULT 0,  
    local_usage TINYINT(1) DEFAULT 1,  
    intl_usage TINYINT(1) DEFAULT 0,  
    card_limit DECIMAL(15,2) DEFAULT 50000.00,  
    role ENUM('user', 'admin') DEFAULT 'user',  
    status ENUM('active', 'frozen') DEFAULT 'active',  
    last_login_date DATE DEFAULT NULL,  
    last_maintenance_claim VARCHAR(7) DEFAULT NULL,  
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP  
);
```

2.2 General Ledger & Trading Tables

```
-- Transactions Log
```

```
CREATE TABLE transactions (  
    id INT AUTO_INCREMENT PRIMARY KEY,  
    account_no VARCHAR(20) NOT NULL,  
    type ENUM('deposit', 'withdrawal', 'transfer') NOT NULL,  
    amount DECIMAL(15, 2) NOT NULL,  
    recipient_account VARCHAR(20) NULL,  
    description VARCHAR(255) NULL,  
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
    FOREIGN KEY (account_no) REFERENCES users(account_no) ON DELETE CASCADE  
);  
  
-- Forex & Crypto Demo Accounts  
CREATE TABLE trading_accounts (  
    id INT AUTO_INCREMENT PRIMARY KEY,  
    user_id INT NOT NULL,  
    asset_class ENUM('forex', 'crypto') NOT NULL,  
    balance DECIMAL(15, 2) DEFAULT 10000.00,  
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
    UNIQUE KEY user_asset (user_id, asset_class),  
    FOREIGN KEY (user_id) REFERENCES users(id) ON DELETE CASCADE  
);
```

2.3 High-Value Transaction MySQL Trigger

```
DELIMITER //  
CREATE TRIGGER after_transaction_insert  
AFTER INSERT ON transactions  
FOR EACH ROW  
BEGIN  
    IF NEW.amount >= 1000.00 THEN  
        UPDATE users  
        SET loyalty_points = loyalty_points + 5  
        WHERE account_no = NEW.account_no;  
    END IF;  
END //  
DELIMITER ;
```

3. CORE CLIENT FEATURES

3.1 Command Hub & Calculator

The dashboard renders real-time financial stats, including account numbers, CNIC profiles, and loyalty states. The available balance can be masked for privacy. It incorporates a client-side currency converter supporting instant calculations across major currencies using client-side JavaScript calculations.

3.2 Wire Transfer & IBFT Portal

- **Internal Wire Transfers:** Dispatches funds instantly to MTAU IBAN profiles. Has zero fees and auto-updates both sender and receiver balance records.
- **Inter-Bank Funds Transfer (IBFT):** Dispatches external wires to major banks. Fees are 0 PKR for Students, 15 PKR for Standard accounts, and 50 PKR for Business accounts.
- **Beneficiary Contact Book:** Allows clients to save recipient profiles to accelerate subsequent transfers.

3.3 Utility Bill Settlements

Surcharge payments for utilities (electricity, water, gas, internet), educational fees, lifestyle memberships, and flight reservations. Selected billers display links to verify routing credentials. Bill payments grant **+25 loyalty points**, while *PIA Airline Flights* grant **+150 loyalty points**.

3.4 Stock & Real Estate Fractional Equity

Invest in fractional growth pools: Clifton Residency shares (9.8% yield), Islamabad Commercial plaza shares (11.2% yield), PSX Tech dividend indexes (15.4% yield), and Energy Giants stocks (13.2% yield).

4. LIVE FOREX & CRYPTO TRADING TERMINAL

A full-featured trading desk simulation allowing users to trade with virtual assets:

- **Desks Activation:** Open Forex and Crypto demo wallets pre-loaded with ****\$10,000 USD**** virtual credits each.
- **TradingView Charts:** Displays live TradingView candlestick widgets.
- **Simulated Margin Rules:**
 - Forex trades execute at ****100x leverage****.
 - Crypto trades execute at ****10x leverage****.
 - Margin checks block trades if the user's free margin is insufficient.
- **Position Tracking:** Monitors open positions, recalculates floating profit/loss, and handles closing actions.
- **Funding Center:**
 - Convert demo profits to checking liquidity at **\$1 USD = 278.40 PKR**.
 - Fund USD demo accounts from checking reserves at **\$1 USD = 278.40 PKR**.

5. AUTOMATED LOYALTY REWARDS & PERKS

Automated rules increase client loyalty points, encouraging user engagement:

- **Daily Login:** Checks `last\_login\_date` to grant ****+1 point**** once per calendar day.
- **Balance Maintenance:** Checks `last\_maintenance\_claim` monthly to grant ****+25 points**** if the user maintains a balance of ****80,000 PKR**** or more.
- **Transaction Reward:** MySQL database trigger grants ****+5 points**** for any transaction $\geq 1,000$ PKR.
- **Perks Catalog:** Redeem points for flight upgrades, restaurant vouchers, or utility bill credits.

6. CONTEXT-AWARE AI CHATBOT ASSISTANT

The floating chatbot assistant parses user queries using client-side NLP. It queries active database records to output customized details on active loans, mutual funds, insurance coverage, and billing dates. State is preserved across pages using `sessionStorage` in the browser.

7. ADMINISTRATIVE CONSOLE

Allows system administrators to override database parameters and manage users:

- **Settings Manager:** Modify system-wide loan interest markup rates and toggle Maintenance Mode.
- **Loan Appraisals:** Review and approve/reject client loan applications. Approvals disburse funds to client accounts and log transactions.
- **User Profile Manager:** Freeze/unfreeze client profiles, reset password credentials to the default `mtau123`, change account classes, or delete profiles.
- **Dispute Desk:** Manage and resolve support tickets submitted by users.
- **Audit Logs:** Filter, search, and audit system activities.

8. CODEBASE DIRECTORY MAPPING

A directory of files in the project workspace:

File Suffix Path	Primary Function
<code>db.php</code>	Establishes the MySQL connection. Included at the top of all backend scripts.
<code>style.css</code>	Global stylesheet containing CSS variables, glassmorphic layout styles, tables, and buttons.
<code>cooleffectslite.js</code>	Interactive canvas particle background animation script.
<code>chatbot.php</code>	Context-aware AI chatbot assistant, included globally on client pages.
<code>dashboard.php</code>	Client dashboard hub containing available balance, card activation, and currency converter.
<code>trading_terminal.php</code>	Forex and crypto demo trading terminal, TradingView charting, and wallet funding.
<code>transfer_terminal.php</code>	Fund transfer console (internal wire, IBFT, and beneficiary management).
<code>bill_terminal.php</code>	Utility bills payment terminal.
<code>mutual_funds_terminal.php</code>	Stock and property fractional profit-sharing terminal.
<code>loan_terminal.php</code>	Loan application console and markup calculations.
<code>manage_card.php</code>	Card management terminal (limits, PIN resets, and lock/freeze controls).
<code>deals_terminal.php</code>	Loyalty rewards deals and perks catalog redemptions.
<code>support_terminal.php</code>	Client support ticket creation.
<code>admin_dashboard.php</code>	Administrative console dashboard for overriding database states and approvals.

File Suffix Path	Primary Function
login.php	Session entry point for clients and administrators.
register.php	User registration with validation and PHPMailer dispatch logic.
statement_pdf.php	Generates a print-friendly statement ledger.
user_manual.php	Print-friendly client user manual.
admin_manual.php	Print-friendly administrator manual.

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